

Series of functions, Differential equations

FMA 2S 007 EP

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Throughout, for a T -periodic piecewise continuous function h the k -th Fourier coefficient is

$$c_k(h) = \frac{1}{T} \int_0^T h(x) e^{-2\pi i k(x/T)} dx, \quad k \in \mathbb{Z}.$$

Throughout this work, we use the following results from Chapter 2 of the textbook:

the one-period identity $\int_a^{a+T} h = \int_0^T h$ (eq. (2.2.4)),

Bessel's inequality (Cor. 2.7.5), Parseval's equality (Thm. 2.7.6),

the criterion which states that "If a function f has its Fourier coefficients $(c_k) \in l^1(\mathbb{Z})$ then $f_N(x) = \sum_{k=-N}^{k=N} c_k e^{2\pi i k(x/T)}$ converges uniformly to a function $f = \sum_{k \in \mathbb{Z}} c_k e^{2\pi i k(x/T)}$, which is continuous and T -periodic." (Prop. 2.4.1),

and the uniqueness of Fourier coefficients (Thm. 2.6.1, Cor. 2.6.2).

Exercise 1

Let $\alpha \in (0, 1]$. We say that a function $f : \mathbb{R} \rightarrow \mathbb{C}$ is α -Hölder if there exists a constant $H_f \geq 0$ such that

$$\forall (x, y) \in \mathbb{R}^2, \quad |f(x) - f(y)| \leq H_f |x - y|^\alpha.$$

1) Give an example of a 1-periodic function which is $(1/2)$ -Hölder but not 1-Hölder.

Proof. We claim that $f(x) = \sqrt{|\sin(\pi x)|}$ has the above property.

Periodicity. Notice that $\sqrt{|\sin(\pi(x+1))|} = \sqrt{|-\sin(\pi x)|} = \sqrt{|\sin(\pi x)|}$. That is, $f(x) = f(x+1)$, meaning the function f is 1-periodic.

(1/2)-Hölder. We will use the following inequality:

$$\left| \sqrt{a} - \sqrt{b} \right| \leq \sqrt{|a - b|} \quad (a, b \geq 0), \quad (1)$$

which follows from

$$(\sqrt{a} - \sqrt{b})^2 = a + b - 2\sqrt{ab} \leq a + b - 2\min(a, b) = |a - b|.$$

For any $x, y \in \mathbb{R}$, applying (??) with $a = |\sin \pi x|$ and $b = |\sin \pi y|$ yields

$$|f(x) - f(y)| \leq \sqrt{||\sin \pi x| - |\sin \pi y||} \leq \sqrt{|\sin \pi x - \sin \pi y|},$$

By the mean value theorem applied to $t \mapsto \sin(\pi t)$, there exists $c \in [\min(x, y), \max(x, y)]$ such that

$$\frac{\sin \pi x - \sin \pi y}{x - y} = (\sin \pi c)' = \pi \cos(\pi c).$$

Since $|\cos(\pi c)| \leq 1$, this gives

$$|\sin \pi x - \sin \pi y| \leq \pi |x - y|.$$

Finally, by monotonicity of the square root, we obtain

$$\boxed{|f(x) - f(y)| \leq \sqrt{\pi} |x - y|^{1/2} .}$$

Hence f is $(1/2)$ -Hölder with $H_f = \sqrt{\pi}$.

Not 1-Hölder. Suppose, for contradiction, that f were 1-Hölder: there would exist $H_f \geq 0$ such that $|f(x) - f(y)| \leq H_f |x - y|$ for all $x, y \in \mathbb{R}$. Taking $y = 0$ and noting $f(0) = \sqrt{|\sin 0|} = 0$, this would force

$$\frac{|f(x) - f(0)|}{|x - 0|} = \frac{\sqrt{|\sin \pi x|}}{|x|} \leq H_f \quad \text{for every } x \neq 0.$$

But, writing

$$\frac{\sqrt{|\sin \pi x|}}{|x|} = \sqrt{\frac{|\sin \pi x|}{\pi |x|} \cdot \frac{\pi}{|x|}}$$

and using $\frac{|\sin \pi x|}{\pi |x|} \rightarrow 1$ while $\frac{\pi}{|x|} \rightarrow +\infty$ as $x \rightarrow 0$,

$$\frac{|f(x) - f(0)|}{|x - 0|} = \frac{\sqrt{|\sin \pi x|}}{|x|} \xrightarrow{x \rightarrow 0} +\infty,$$

which contradicts the bound $\leq H_f$. Hence f is not 1-Hölder. $\square$

From now on f is a fixed 1-periodic α -Hölder function (so $T = 1$), and for $t \in \mathbb{R}$ we set $g_t : \mathbb{R} \ni x \mapsto f(x - t) - f(x) \in \mathbb{C}$.

2.a) Show that g_t is a continuous 1-periodic function that satisfies $\sup_{x \in \mathbb{R}} |g_t(x)| \leq H_f |t|^\alpha$.

Proof. Being α -Hölder, f is uniformly continuous, hence so is g_t as a sum of uniformly continuous functions.

Notice that, since f is 1-periodic,

$$g_t(x + 1) = f(x + 1 - t) - f(x + 1) = f(x - t) - f(x) = g_t(x),$$

Hence, g_t is 1-periodic.

Finally, using the Hölder property of f , for every x ,

$$|g_t(x)| = |f(x - t) - f(x)| \leq H_f |(x - t) - x|^\alpha = H_f |t|^\alpha,$$

so $\sup_{x \in \mathbb{R}} |g_t(x)| \leq H_f |t|^\alpha$. $\square$

2.b) Compute the Fourier coefficients of g_t .

Proof. Recall that, since g_t and f are 1-periodic (here $T = 1$), their Fourier coefficients are by definition

$$c_k(g_t) = \int_0^1 g_t(x) e^{-2\pi i k x} dx, \quad c_k(f) = \int_0^1 f(x) e^{-2\pi i k x} dx.$$

Substituting the definition $g_t(x) = f(x - t) - f(x)$ and splitting the integral by linearity,

$$c_k(g_t) = \int_0^1 (f(x - t) - f(x)) e^{-2\pi i k x} dx = \underbrace{\int_0^1 f(x - t) e^{-2\pi i k x} dx}_{=: I} - \underbrace{\int_0^1 f(x) e^{-2\pi i k x} dx}_{= c_k(f)}.$$

It remains to evaluate I . Using the change of variable $u = x - t$, $du = dx$ and the bounds $x = 0, 1$ become $u = -t, 1 - t$:

$$I = \int_{-t}^{1-t} f(u) e^{-2\pi i k (u+t)} du = e^{-2\pi i k t} \int_{-t}^{1-t} f(u) e^{-2\pi i k u} du,$$

where the factor $e^{-2\pi ikt}$ is constant in u and the map $u \mapsto f(u) e^{-2\pi iku}$ is 1-periodic (as a product of the 1-periodic f and the 1-periodic $u \mapsto e^{-2\pi iku}$), so by the one-period identity (2.2.4) its integral over the interval $[-t, 1-t]$ of length 1 equals its integral over $[0, 1]$. Hence,

$$I = e^{-2\pi ikt} \int_{-t}^{1-t} f(u) e^{-2\pi iku} du = e^{-2\pi ikt} \int_0^1 f(u) e^{-2\pi iku} du = e^{-2\pi ikt} c_k(f).$$

Hence $I = e^{-2\pi ikt} c_k(f)$, and therefore

$$c_k(g_t) = I - c_k(f) = e^{-2\pi ikt} c_k(f) - c_k(f) = (e^{-2\pi ikt} - 1) c_k(f). \quad \square$$

2.c) Show that $|c_k(g_{1/(2k)})| = 2 |c_k(f)|$.

Proof. Take $t = \frac{1}{2k}$ in the formula of 2.b. Then $e^{-2\pi ik \cdot \frac{1}{2k}} = e^{-i\pi} = -1$, so

$$c_k(g_{1/(2k)}) = (e^{-i\pi} - 1) c_k(f) = (-1 - 1) c_k(f) = -2 c_k(f),$$

By taking the modulus on both sides, $|c_k(g_{1/(2k)})| = 2 |c_k(f)|$. $\square$

3) Prove that there exists $C_\alpha > 0$ such that $\forall k \in \mathbb{Z}^*$, $|c_k(f)| \leq C_\alpha \mathbf{H}_f |k|^{-\alpha}$.

Proof. By 2.c, taking moduli in $c_k(g_{1/(2k)}) = -2 c_k(f)$ gives

$$|c_k(f)| = \frac{1}{2} |c_k(g_{1/(2k)})|.$$

Now, we write out the coefficient $c_k(g_{1/(2k)})$ from its definition, and then estimate. By definition,

$$c_k(g_{1/(2k)}) = \int_0^1 g_{1/(2k)}(x) e^{-2\pi i k x} dx.$$

Hence, taking the modulus inside the integral and using $|e^{-2\pi i k x}| = 1$,

$$|c_k(g_{1/(2k)})| = \left| \int_0^1 g_{1/(2k)}(x) e^{-2\pi i k x} dx \right| \leq \int_0^1 |g_{1/(2k)}(x)| |e^{-2\pi i k x}| dx = \int_0^1 |g_{1/(2k)}(x)| dx.$$

Bounding the integrand by its supremum and applying 2.a with $t = \frac{1}{2k}$,

$$\int_0^1 |g_{1/(2k)}(x)| dx \leq \sup_{x \in \mathbb{R}} |g_{1/(2k)}(x)| \stackrel{2.a}{\leq} \mathbf{H}_f \left(\frac{1}{2|k|} \right)^\alpha.$$

Combining the three displays,

$$|c_k(f)| = \frac{1}{2} |c_k(g_{1/(2k)})| \leq \frac{1}{2} \mathbf{H}_f \left(\frac{1}{2|k|} \right)^\alpha = 2^{-(\alpha+1)} \mathbf{H}_f |k|^{-\alpha},$$

so the statement holds with $C_\alpha = 2^{-(\alpha+1)}$. $\square$

For the remaining parts, fix $n \in \mathbb{N}$ and put $t_n = (3 \cdot 2^n)^{-1}$.

4.a) Explain why $\sum_{2^n \leq k < 2^{n+1}} |c_k(g_{t_n})|^2 \leq \int_0^1 |g_{t_n}(x)|^2 dx$.

Proof. g_{t_n} is continuous and 1-periodic (by 2.a), so Parseval's equality (Thm. 2.7.6) gives

$$\int_0^1 |g_{t_n}(x)|^2 dx = \sum_{k \in \mathbb{Z}} |c_k(g_{t_n})|^2.$$

The sum on the left of the statement runs over the *subset* $\{2^n \leq k < 2^{n+1}\}$ of $\mathbb{Z}$, and all terms are ≥ 0 ; therefore it is bounded by the total sum, which equals $\int_0^1 |g_{t_n}|^2$. (Equivalently, this is Bessel's inequality, Cor. 2.7.5.) $\square$

4.b) Deduce that $\sum_{2^n \leq k < 2^{n+1}} |c_k(f)|^2 \leq 3^{-1} \left(\frac{1}{3 \cdot 2^n} \right)^{2\alpha} H_f^2$.

[One can prove and use that for any $k \in [2^n, 2^{n+1})$ one has $|e^{-2\pi i k t_n} - 1| \geq \sqrt{3}$.]

Proof. The bracketed inequality. For any real θ , $|e^{-i\theta} - 1|^2 = (\cos \theta - 1)^2 + \sin^2 \theta = 2 - 2 \cos \theta = 4 \sin^2(\theta/2)$, so $|e^{-i\theta} - 1| = 2 |\sin(\theta/2)|$. Take $\theta = 2\pi k t_n = \frac{2\pi}{3} \cdot \frac{k}{2^n}$. If $2^n \leq k < 2^{n+1}$ then $\frac{k}{2^n} \in [1, 2)$, hence $\theta \in [\frac{2\pi}{3}, \frac{4\pi}{3})$ and $\frac{\theta}{2} \in [\frac{\pi}{3}, \frac{2\pi}{3})$. On this interval $\sin$ increases on $[\pi/3, \pi/2]$ and decreases on $[\pi/2, 2\pi/3]$, with equal endpoint values $\sin(\pi/3) = \sin(2\pi/3) = \frac{\sqrt{3}}{2}$; hence $\sin(\theta/2) \geq \frac{\sqrt{3}}{2}$ and

$$|e^{-2\pi i k t_n} - 1| = 2 \sin(\theta/2) \geq \sqrt{3} \quad (2^n \leq k < 2^{n+1}).$$

(For $-2^{n+1} < k \leq -2^n$ one has $\theta/2 \in (-\frac{2\pi}{3}, -\frac{\pi}{3}]$, where $|\sin(\theta/2)| \geq \frac{\sqrt{3}}{2}$ as well, so the same bound holds.)

The estimate. By 2.b, $|c_k(g_{t_n})|^2 = |e^{-2\pi i k t_n} - 1|^2 |c_k(f)|^2 \geq 3 |c_k(f)|^2$ on the block, i.e. $|c_k(f)|^2 \leq \frac{1}{3} |c_k(g_{t_n})|^2$. Summing and using 4.a, then 2.a,

$$\sum_{2^n \leq k < 2^{n+1}} |c_k(f)|^2 \leq \frac{1}{3} \sum_{2^n \leq k < 2^{n+1}} |c_k(g_{t_n})|^2 \leq \frac{1}{3} \int_0^1 |g_{t_n}(x)|^2 dx \leq \frac{1}{3} \left(\sup_x |g_{t_n}| \right)^2 \leq \frac{1}{3} H_f^2 t_n^{2\alpha},$$

which is exactly $3^{-1} \left(\frac{1}{3 \cdot 2^n} \right)^{2\alpha} H_f^2$. $\square$

4.c) Deduce that there exists a constant C'_α such that $\sum_{2^n \leq k < 2^{n+1}} |c_k(f)| \leq C'_\alpha H_f 2^{(\frac{1}{2}-\alpha)n}$.

Proof. The block $\{k : 2^n \leq k < 2^{n+1}\}$ contains exactly $2^{n+1} - 2^n = 2^n$ integers. By Cauchy-Schwarz,

$$\sum_{2^n \leq k < 2^{n+1}} |c_k(f)| = \sum_{2^n \leq k < 2^{n+1}} |c_k(f)| \cdot 1 \leq \left(\sum_{2^n \leq k < 2^{n+1}} |c_k(f)|^2 \right)^{1/2} \left(\sum_{2^n \leq k < 2^{n+1}} 1^2 \right)^{1/2}.$$

The second factor is $(2^n)^{1/2} = 2^{n/2}$. By inserting the bound of 4.b for the first factor, we get:

$$\sum_{2^n \leq k < 2^{n+1}} |c_k(f)| \leq 2^{n/2} \left(3^{-1} (3 \cdot 2^n)^{-2\alpha} H_f^2 \right)^{1/2} = 2^{n/2} \cdot 3^{-1/2} 3^{-\alpha} 2^{-\alpha n} H_f = 3^{-(\alpha+\frac{1}{2})} H_f 2^{(\frac{1}{2}-\alpha)n},$$

so the statement holds with $C'_\alpha = 3^{-(\alpha+1/2)}$. $\square$

4.d) Prove that if $\alpha \in (1/2, 1]$, the Fourier series of f converges normally to f .

Proof. Summability of the coefficients. Decompose $\mathbb{Z}^*$ into dyadic blocks. For the positive indices,

$$\sum_{k \geq 1} |c_k(f)| = \sum_{n=0}^{\infty} \sum_{2^n \leq k < 2^{n+1}} |c_k(f)| \stackrel{4.c}{\leq} \sum_{n=0}^{\infty} C'_\alpha \text{H}_f 2^{\left(\frac{1}{2}-\alpha\right)n} = C'_\alpha \text{H}_f \sum_{n=0}^{\infty} \left(2^{\frac{1}{2}-\alpha}\right)^n.$$

Since $\alpha > \frac{1}{2}$ we have $\frac{1}{2} - \alpha < 0$, so $2^{\frac{1}{2}-\alpha} < 1$ and the geometric series converges:

$$\sum_{k \geq 1} |c_k(f)| \leq \frac{C'_\alpha \text{H}_f}{1 - 2^{\frac{1}{2}-\alpha}} < \infty.$$

Using the negative-index blocks $\{-2^{n+1} < k \leq -2^n\}$ together with the second case of the bracketed inequality in 4.b, the estimates of 4.b and 4.c hold verbatim there, hence $\sum_{k \leq -1} |c_k(f)| < \infty$ as well. Adding the (finite) term $|c_0(f)|$,

$$\sum_{k \in \mathbb{Z}} |c_k(f)| < \infty, \quad \text{i.e. } (c_k(f))_{k \in \mathbb{Z}} \in \ell^1(\mathbb{Z}).$$

Convergence and identification of the limit. By Proposition 2.4.1, the series $\sum_{k \in \mathbb{Z}} c_k(f) e^{2\pi i k x}$ converges *normally* (and uniformly) to a continuous 1-periodic function g , whose Fourier coefficients are $c_k(g) = c_k(f)$ for all k . The function $f - g$ is continuous, 1-periodic and has all Fourier coefficients equal to 0; by the uniqueness theorem (Thm. 2.6.1 / Cor. 2.6.2) it is identically zero. Therefore $g = f$, i.e.

$$f(x) = \sum_{k \in \mathbb{Z}} c_k(f) e^{2\pi i k x} \quad \text{normally on } \mathbb{R}. \quad \square$$

Exercise 2

Let $\alpha \in \mathbb{R} \setminus \mathbb{Z}$ and

$$f_\alpha : [0, 2\pi] \ni x \mapsto \frac{\pi}{\sin(\pi\alpha)} e^{i(\pi-x)\alpha} \in \mathbb{C}.$$

We denote by $\tilde{f}_\alpha$ the 2π -periodization of f_α (so here $T = 2\pi$). Since f_α is C^∞ on $[0, 2\pi]$, $\tilde{f}_\alpha$ is piecewise continuous (with at worst a jump at the points $2\pi\mathbb{Z}$), so the Fourier theory of Chapter 2 applies.

1) Compute the Fourier coefficients of $\tilde{f}_\alpha$.

Proof. With $T = 2\pi$, the definition of the Fourier coefficients is: $c_k(\tilde{f}_\alpha) = \frac{1}{2\pi} \int_0^{2\pi} f_\alpha(x) e^{-ikx} dx$.

Thus:

$$\begin{aligned} c_n(\tilde{f}_\alpha) &= \frac{1}{2\pi} \int_0^{2\pi} f_\alpha(x) e^{-inx} dx = \frac{1}{2\pi} \cdot \frac{\pi}{\sin(\pi\alpha)} \cdot e^{i\pi\alpha} \int_0^{2\pi} e^{-ix(n+\alpha)} dx \\ c_n(\tilde{f}_\alpha) &= \frac{e^{i\pi\alpha}}{2\sin(\pi\alpha)} \int_0^{2\pi} e^{-i(n+\alpha)x} dx \end{aligned}$$

Now, we compute the integral $\int_0^{2\pi} e^{-i(n+\alpha)x} dx$

$$\alpha \notin \mathbb{Z} \implies \alpha + n \neq 0 \quad \forall n \in \mathbb{Z}$$

$$\begin{aligned}
\int_0^{2\pi} e^{-i(n+\alpha)x} dx &= \left[\frac{e^{-i(n+\alpha)x}}{-i(n+\alpha)} \right]_0^{2\pi} = \frac{e^{-i(n+\alpha)2\pi} - 1}{-i(\alpha+n)} = \\
&= \frac{e^{-2\pi i n} e^{-2\pi i \alpha} - 1}{-i(\alpha+n)} = \frac{e^{-2\pi i \alpha} - 1}{-i(\alpha+n)} = \frac{e^{-i\pi\alpha}(e^{-i\pi\alpha} - e^{i\pi\alpha})}{-i(\alpha+n)} = \\
&= \frac{e^{-i\pi\alpha}}{-i(\alpha+n)} [-2i \sin(\pi\alpha)] = \frac{2e^{-i\pi\alpha} \sin(\pi\alpha)}{\alpha+n}
\end{aligned}$$

Hence, we get:

$$c_n(\tilde{f}_\alpha) = \frac{e^{i\pi\alpha}}{2 \sin(\pi\alpha)} \cdot \frac{2 \sin(\pi\alpha) \cdot e^{-i\pi\alpha}}{\alpha+n} = \frac{1}{\alpha+n}$$

$$\boxed{c_n(\tilde{f}_\alpha) = \frac{1}{\alpha+n}}$$

□

2) Show that $\sum_{k \in \mathbb{Z}} \frac{1}{(k+\alpha)^2} = \frac{\pi^2}{(\sin(\pi\alpha))^2}$.

Proof. f_α is continuous on $[0, 2\pi]$ since it is a composition of continuous functions

$$\alpha \notin \mathbb{Z} \implies \sin(\pi\alpha) \neq 0$$

Therefore, $\tilde{f}_\alpha$ is piecewise continuous on $\mathbb{R}$

Hence,

$$\tilde{f}_\alpha \in L^2([0, 2\pi])$$

We can use Parseval's identity:

$$\sum_{n \in \mathbb{Z}} |c_n(\tilde{f}_\alpha)|^2 = \frac{1}{2\pi} \int_0^{2\pi} |\tilde{f}_\alpha(x)|^2 dx$$

From 1) :

$$\sum_{n \in \mathbb{Z}} |c_n(\tilde{f}_\alpha)|^2 = \sum_{n \in \mathbb{Z}} \frac{1}{(n+\alpha)^2}$$

$$\begin{aligned}
\frac{1}{2\pi} \int_0^{2\pi} |\tilde{f}_\alpha(x)|^2 dx &= \frac{1}{2\pi} \int_0^{2\pi} \left| \frac{\pi}{\sin(\pi\alpha)} e^{i(\pi-x)\alpha} \right|^2 dx = \\
&= \frac{1}{2\pi} \cdot \frac{\pi^2}{\sin^2(\pi\alpha)} \int_0^{2\pi} |e^{i(\pi-x)\alpha}|^2 dx
\end{aligned}$$

$$(\pi-x)\alpha \in \mathbb{R} \implies |e^{i(\pi-x)\alpha}| = 1$$

Hence, we get :

$$\frac{1}{2\pi} \cdot \frac{\pi^2}{\sin^2(\pi\alpha)} \int_0^{2\pi} 1 dx = \frac{1}{2\pi} \cdot \frac{\pi^2}{\sin^2(\pi\alpha)} \cdot 2\pi = \frac{\pi^2}{\sin^2(\pi\alpha)}.$$

□